Crucible — Final Project Outline (v8)

UiPath AgentHack · Track 3: Test Cloud

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1. Elevator Pitch

> Crucible is a multi-agent testing intelligence platform that watches your pull requests, generates production-quality tests with an adversarial writer/critic loop, runs a security scan, executes the tests locally, and ships JUnit XML results into UiPath Test Cloud — then reads the actual pass/fail data back to close the feedback loop. Every test is traceable to a requirement. Every flaky test is flagged before it merges.

One-line Devpost: *Three agents, one adversarial loop, bidirectional Test Cloud integration — confidence scores grounded in real execution results, not just static analysis.*

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2. What Makes It Unique

Differentiator	Why it matters
Adversarial writer/critic loop	Tests hammered until a critic approves — not generated once
Bidirectional TC integration	Crucible *sends* results AND *reads back* pass/fail to adjust the confidence score
JUnit XML as TC contract	pytest runs locally → XML exported → posted to TC (format TC actually accepts)
Async pipeline with queue	Webhook server never blocks — `asyncio.Queue` worker handles concurrent PRs
Semgrep security layer	OWASP Top 10, secrets, Python antipatterns — warning cap separates severity levels
Requirements traceability	Every test carries `req_id` linked to source requirement — auditable
FRAGILE/FLAKY detection	Static heuristics flag time-dependent, order-dependent, hardcoded-port tests
HMAC-verified webhook	A security project that verifies its own webhook — internally consistent
Live confidence score	0–1 grounded in actual TC pass rate with graceful fallback if TC unavailable

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3. Architecture



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4. Webhook Handler (Async, HMAC-verified)

```
from fastapi import FastAPI, Request, HTTPException
import asyncio, hmac, hashlib, httpx, json, os
```

```
app = FastAPI()
WEBHOOK_SECRET = os.environ["GITHUB_WEBHOOK_SECRET"]
pipeline_queue: asyncio.Queue = asyncio.Queue()

@app.on_event("startup")
async def start_worker():
    asyncio.create_task(pipeline_worker())

async def pipeline_worker():
    """Single background worker — processes one PR at a time, never blocks event
loop."""
    while True:
        item = await pipeline_queue.get()
        try:
            await run_crucible_pipeline(**item)
        except Exception as e:
            log_error(e)
        finally:
            pipeline_queue.task_done()

@app.post("/webhook")
async def github_webhook(request: Request):
    body = await request.body()
    sig = request.headers.get("X-Hub-Signature-256", "")
    if not verify_signature(body, sig):
        raise HTTPException(status_code=403, detail="Invalid signature")

    payload = json.loads(body)
    if payload.get("action") not in ("opened", "synchronize"):
        return {"status": "ignored"}

    pr_number = payload["pull_request"]["number"]
    # changed_files is an INTEGER in the payload — must call files API separately:
    async with httpx.AsyncClient() as client:
        resp = await client.get(
            f"https://api.github.com/repos/{OWNER}/{REPO}/pulls/{pr_number}/files",
            headers={"Authorization": f"Bearer {GITHUB_TOKEN}"}
        )
    files = resp.json()  # array of {filename, patch, status}

    await pipeline_queue.put({"files": files, "pr_number": pr_number})
    return {"status": "accepted"}
```

HMAC verification (positional args, unit-tested):

```
def verify_signature(payload_body: bytes, signature_header: str) -> bool:
    if not signature_header or not signature_header.startswith("sha256="):
        return False
    expected = "sha256=" + hmac.new(
        WEBHOOK_SECRET.encode(),
        payload_body,       # positional — not keyword — avoids Python version issues
        hashlib.sha256
    ).hexdigest()
```

```
    return hmac.compare_digest(expected, signature_header)

# Unit tests — run on Day 5 before integration:
def test_hmac_valid():
    secret = "test_secret"
    payload = b'{"action": "opened"}'
    sig = "sha256=" + hmac.new(secret.encode(), payload, hashlib.sha256).hexdigest()
    assert verify_signature(payload, sig) is True

def test_hmac_bad_hash():
    assert verify_signature(b"payload", "sha256=badhash") is False

def test_hmac_empty_header():
    assert verify_signature(b"payload", "") is False
```

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5. Agents in Detail

5a. Requirements Analyzer (claude-haiku-4-5-20251001)

- **Input:** PR description, docstrings, linked ticket text
- **Output:** `[{ "req_id": "REQ-042", "description": "...", "priority": "HIGH" }]`
- **Fallback:** No requirements found → `req_id: "UNLINKED"` + warning in PR comment

5b. Writer Agent (claude-sonnet-4-6)

- **Input:** PR diff, req_id list, previous iteration feedback
- **Output:** Raw pytest file — no markdown, no fences, no explanation
- **Tags each test:** `@pytest.mark.req_id("REQ-042")`
- **Iteration 1 prompt:** `"Write pytest tests for this diff. Tag each with @pytest.mark.req_id(). Return only raw Python."`
- **Iteration 2+ prompt:** `"Fix these issues: {feedback}. Previous attempt: {code}. Return only raw Python."`

5c. Critic Agent (claude-sonnet-4-6)

- **Output:** `APPROVED` or `ISSUES:\n- <specific issue>`
- **Checks:** Missing edge cases, wrong assertions, missing teardown, non-determinism, missing parametrize
- **Max iterations:** 5

5d. Security Agent (claude-sonnet-4-6 + Semgrep) — tempfile + hash cache

```
import tempfile, os, hashlib, json, asyncio

security_cache: dict = {"file_hash": None, "findings": None, "score": None}
```

```
async def run_security_agent(test_file_content: str) -> tuple[list, float]:
    # Hash first — use as cache key and for dedup
    current_hash = hashlib.sha256(test_file_content.encode()).hexdigest()
    if security_cache["file_hash"] == current_hash:
        return security_cache["findings"], security_cache["score"]  # cache hit

    # Semgrep scans files/directories — NOT stdin source directly.
    # Write to a named tempfile, scan it, then clean up.
    with tempfile.NamedTemporaryFile(suffix=".py", delete=False, mode="w") as f:
        f.write(test_file_content)
        tmp_path = f.name

    try:
        proc = await asyncio.create_subprocess_exec(
            "semgrep",
            "--config", "p/python",
            "--config", "p/owasp-top-ten",
            "--config", "p/secrets",
            "--json", tmp_path,          # file path — not "-"
            stdout=asyncio.subprocess.PIPE,
            stderr=asyncio.subprocess.PIPE
        )
        stdout, _ = await proc.communicate()
        findings = json.loads(stdout).get("results", [])
    finally:
        os.unlink(tmp_path)  # always clean up

    score = compute_security_score(findings)
    security_cache.update({
        "file_hash": current_hash,
        "findings": findings,
        "score": score
    })
    return findings, score
```

5e. Regression Impact Agent (claude-sonnet-4-6) — optional

- **Mode A:** Greenfield — establishes coverage baseline
- **Mode B:** Existing project — compares vs. stored baseline, flags regressions
- **Graceful degradation:** If cut due to time, pipeline skips silently — confidence formula unchanged

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6. Confidence Score (Final — All Bugs Fixed)

```
def compute_confidence(
    iteration: int,
    warnings: list,
    errors: list,
    any_critical: bool,
    tc_pass_rate: float | None
```

```
) -> tuple[float, str]:

    # critic_score: continuous, floored at 0.5, no special case needed
    critic_score = max(0.5, 1.1 - 0.1 * iteration)
    # iteration 1→1.0, 2→0.9, 3→0.8, 4→0.7, 5→0.6, floor 0.5

    # security_score: cap warnings at 4 to prevent conflating with CRITICAL
    warning_penalty = min(len(warnings), 4) * 0.15   # max penalty = 0.60
    error_penalty   = len(errors) * 0.5
    security_score  = 0.0 if any_critical else max(0.0, 1.0 - warning_penalty -
error_penalty)

    # tc_pass_rate: two formula modes
    if tc_pass_rate is not None:
        confidence = 0.3 * critic_score + 0.3 * security_score + 0.4 * tc_pass_rate
        tc_label = f"{tc_pass_rate:.0%} TC pass rate"
    else:
        # Renormalize to 60/40 — TC term excluded
        confidence = 0.6 * critic_score + 0.4 * security_score
        tc_label = "⚠️ TC Pending — score excludes execution data"

    return round(confidence, 2), tc_label
```

Security score table:

Findings	`security_score`
0 findings	1.00
1 warning	0.85
2 warnings	0.70
3 warnings	0.55
4+ warnings	**0.40** ← floor for warnings alone
1 error	0.50
Any CRITICAL	**0.00** ← unambiguous, never confused with warnings

Merge thresholds:

- `≥ 0.85` → ✅ Auto-approve
- `0.65–0.84` → ⚠️ Review recommended
- `< 0.65` → ❌ Block merge

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7. Test Cloud Integration (Standalone First)

Build `tc_client.py` as a fully standalone, independently testable module on **Day 2** — before any agent code is written.

```
TC_DRY_RUN = os.environ.get("TC_DRY_RUN", "false").lower() == "true"
```

```
async def full_tc_roundtrip(xml_path: str) -> float | None:
    """Upload JUnit XML → trigger run → poll → return pass rate. None on failure."""
    if TC_DRY_RUN:
        print("[TC_DRY_RUN] Stubbing tc_pass_rate = 0.78")
        return 0.78

    async with httpx.AsyncClient(timeout=30.0) as client:
        # Step 1: Upload
        with open(xml_path, "rb") as f:
            upload = await client.post(
                f"{TC_BASE}/api/v2/projects/{TC_PROJECT_ID}/test-runs/results",
                headers={"Authorization": f"Bearer {TC_TOKEN}"},
                files={"file": ("results.xml", f, "application/xml")}
            )
        if upload.status_code not in (200, 201):
            print(f"[TC] Upload failed: {upload.status_code} {upload.text}")
            return None

        # Step 2: Trigger
        trigger = await client.post(
            f"{TC_BASE}/api/v2/projects/{TC_PROJECT_ID}/test-runs",
            headers={"Authorization": f"Bearer {TC_TOKEN}"},
            json={}
        )
        run_id = trigger.json().get("id")

        # Step 3: Poll
        for _ in range(30):  # max ~5 min
            await asyncio.sleep(10)
            status = await client.get(
                f"{TC_BASE}/api/v2/projects/{TC_PROJECT_ID}/test-runs/{run_id}",
                headers={"Authorization": f"Bearer {TC_TOKEN}"}
            )
            data = status.json()
            if data.get("status") != "running":
                passed = data.get("passedCount", 0)
                total  = data.get("totalCount", 1)
                return passed / total if total > 0 else None

    return None  # timeout
```

tc_pass_rate state handling:

State	Value	PR comment label
TC run completed	`passed / total`	`25% TC pass rate`
TC API timeout	`None`	⚠️ `TC Pending — renormalized score`
TC auth failure	`None`	⚠️ `TC Error — check credentials`
`TC_DRY_RUN=true`	`0.78` (stub)	`(dry run)`

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8. Shared State: SQLite

Why SQLite: One file, zero extra services in docker-compose, both FastAPI and Streamlit read/write it, 500ms polling latency is fine for a demo.

```
# state.py — imported by both FastAPI and Streamlit
import sqlite3, os
from pathlib import Path

DB_PATH = Path(os.environ.get("STATE_DB", "/tmp/crucible_state.db"))

def init_db():
    with sqlite3.connect(DB_PATH) as conn:
        conn.execute("""
            CREATE TABLE IF NOT EXISTS pipeline_runs (
                pr_number    INTEGER PRIMARY KEY,
                step         INTEGER DEFAULT 0,
                step_label   TEXT    DEFAULT 'Queued',
                confidence   REAL,
                tc_pass_rate REAL,
                findings     TEXT,   -- JSON array
                req_coverage TEXT,   -- JSON object
                fragile_count INTEGER DEFAULT 0,
                updated_at   TIMESTAMP DEFAULT CURRENT_TIMESTAMP
            )
        """)

def update_step(pr_number: int, step: int, label: str):
    with sqlite3.connect(DB_PATH) as conn:
        conn.execute("""
            INSERT INTO pipeline_runs (pr_number, step, step_label, updated_at)
            VALUES (?, ?, ?, CURRENT_TIMESTAMP)
            ON CONFLICT(pr_number) DO UPDATE SET
                step=excluded.step,
                step_label=excluded.step_label,
                updated_at=CURRENT_TIMESTAMP
        """, (pr_number, step, label))

def get_run(pr_number: int) -> dict | None:
    with sqlite3.connect(DB_PATH) as conn:
        conn.row_factory = sqlite3.Row
        row = conn.execute(
            "SELECT * FROM pipeline_runs WHERE pr_number = ?", (pr_number,)
        ).fetchone()
    return dict(row) if row else None
```

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9. FRAGILE/FLAKY Detection

```
import re

FRAGILE_PATTERNS = [
```

```
    r"time\.sleep\(",
    r"datetime\.now\(",
    r"random\.",            # without seed
    r"localhost:\d{4,5}",   # hardcoded ports
    r"global\s+\w+",        # global state mutation
    r"os\.environ\[",       # env reads inside test body
]

def is_fragile(test_source: str) -> bool:
    return any(re.search(p, test_source) for p in FRAGILE_PATTERNS)
```

Naming strategy:

- **Preferred:** Custom property `{"name": "FRAGILE", "value": true}` on TC test case
- **Fallback:** `[FRAGILE]` prefix in test name if custom properties return 422
- Both paths confirmed in `validate_apis.py` on Day 1

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10. Streamlit Dashboard

```
PIPELINE_STEPS = [
    "Webhook received & verified",
    "PR files fetched",
    "Requirements analyzed",
    "Tests generated (Writer)",
    "Tests reviewed (Critic)",
    "Security scanned (Semgrep)",
    "pytest executed",
    "Results uploaded to TC",
    "TC run polled & scored",
]

# Sidebar — keeps audience oriented during TC polling phase
st.sidebar.header("Pipeline State")
for i, step_name in enumerate(PIPELINE_STEPS, 1):
    if i < current_step:
        st.sidebar.write(f"✅ {step_name}")
    elif i == current_step:
        st.sidebar.write(f"🔄 **[STEP {i}/{len(PIPELINE_STEPS)}: {step_name}...]**")
    else:
        st.sidebar.write(f"⬜ {step_name}")

# Main area
col1, col2 = st.columns(2)
with col1:
    st.metric("Confidence Score", f"{confidence:.2f}", delta=tc_label)
    st.progress(coverage_pct / 100, text=f"Coverage: {coverage_pct}%")
with col2:
    st.dataframe(findings_df)       # severity / rule_id / line
    st.dataframe(req_coverage_df)   # req_id / test_count / tc_pass_fail
```

Streamlit polls SQLite every 500ms — no pub/sub needed for a demo.

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11. Day 1 Validation Checklist

```
python validate_apis.py
```

Run this before writing any agent code. All 7 must pass:

```
✅ 1. JUnit XML upload → TC /test-runs/results → 200/201
✅ 2. GET /test-runs → schema confirmed, passedCount/totalCount fields located
✅ 3. POST /test-runs trigger → run starts, run_id returned
✅ 4. GET /test-runs/{id} poll → status field values documented ("running", "passed", etc.)
✅ 5. FRAGILE custom property → confirmed OR fallback [FRAGILE] prefix documented
✅ 6. Maestro Case POST → confirmed OR "not available, skipping" logged
✅ 7. semgrep --validate --config p/python → exit code 0
```

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12. Environment Variables

```
# .env.example — fully annotated

# Anthropic
ANTHROPIC_API_KEY=sk-ant-...

# GitHub
GITHUB_TOKEN=ghp_...
GITHUB_WEBHOOK_SECRET=your_webhook_secret    # must match GitHub repo webhook settings
GITHUB_OWNER=your_org
GITHUB_REPO=your_repo

# UiPath Test Cloud
TC_BASE_URL=https://cloud.uipath.com/{org}/{tenant}
TC_TOKEN=your_tc_bearer_token
TC_PROJECT_ID=your_project_id

# UiPath Maestro (optional — pipeline skips gracefully if unset)
MAESTRO_BASE_URL=https://cloud.uipath.com/{org}/{tenant}/orchestrator_
MAESTRO_TOKEN=your_maestro_token

# Development flags
TC_DRY_RUN=false      # "true" → stubs tc_pass_rate=0.78, no TC API calls
STATE_DB=/tmp/crucible_state.db   # SQLite path shared between FastAPI and Streamlit
```

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13. Build Timeline

Week 1

Day 1 — Validate everything first

Run `validate_apis.py` — all 7 checks must pass before any agent code is written.

Day 2 — TC standalone mini-milestone

- Build `tc_client.py` with `full_tc_roundtrip()` end-to-end
- Test with a real JUnit XML (generate from any local pytest run)
- Confirm pass rate extraction from actual TC response schema
- Confirm `TC_DRY_RUN=true` stub path works
- *No agents yet — TC pipeline fully standalone and testable*

Days 3–4 — Agent code

- Requirements Analyzer (haiku)
- Writer + Critic adversarial loop (port from `code_pipeline.py`)
- Security agent: tempfile → semgrep → sha256 cache → Claude summary
- Confidence formula wired to `tc_client.py`
- SQLite state writes at each step

Day 5 — Webhook + end-to-end

- FastAPI webhook handler: HMAC verified, `asyncio.Queue` worker
- HMAC unit tests (three cases: valid / bad hash / empty header)
- `asyncio.create_subprocess_exec` for pytest
- `httpx.AsyncClient` throughout — no `requests.post` anywhere
- PR comment formatter (handles `tc_pass_rate = None` gracefully)
- End-to-end smoke test on a real PR

Week 2

Days 6–7 — Demo target + dashboard

- `vulnerable_fixture.py`: known bugs, 1 CRITICAL Semgrep hit, 2 FRAGILE patterns, 3 linked req_ids, `tc_pass_rate = 0.25` in staged payload
- FRAGILE detection heuristics wired into pipeline
- Streamlit dashboard with SQLite polling + pipeline step indicator
- Regression Impact Agent (greenfield mode — optional, never blocks pipeline)

Days 8–9 — Containerize

```
# docker-compose.yml
services:
  webhook:
    build: .
    command: uvicorn main:app --host 0.0.0.0 --port 8000
    env_file: .env
    volumes:
```

```
      - crucible_state:/tmp

  dashboard:
    build: .
    command: streamlit run dashboard.py --server.port 8501
    env_file: .env
    volumes:
      - crucible_state:/tmp

volumes:
  crucible_state:   # shared SQLite lives here
```

- Full test on clean `docker-compose up`
- Annotated `.env.example` finalized

Day 10 — Ship

- Demo rehearsal with pre-staged payload
- Devpost write-up
- README: "Built with Claude Code" badge
- Final smoke test on clean clone

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14. Demo Plan

Setup (T−30s before opening demo slide):

- `vulnerable_fixture.py` payload queued — Writer will need exactly 2 iterations to approve
- `TC_DRY_RUN=false` — real TC integration live
- Streamlit open at Step 1

3-minute flow:

Step	What happens	Dashboard shows
1	Webhook fires → HMAC verified	✅ Step 1/9
2	PR files fetched via API	✅ Step 2/9
3	Requirements analyzed → 3 req_ids	✅ Step 3/9
4	Writer iteration 1 → Critic rejects	🔄 Step 4/9
5	Writer iteration 2 → Critic APPROVES	✅ Step 5/9
6	Security scan: 1 CRITICAL, 2 warnings (cache miss — new content)	✅ Step 6/9
7	pytest runs → JUnit XML generated	✅ Step 7/9
8	XML uploaded to TC → run triggered	✅ Step 8/9
9	TC polled → 25% pass rate returned	✅ Step 9/9

Confidence score (verified against formula):

```
critic_score    = max(0.5, 1.1 - 0.1×2) = 0.9
security_score = 0.0    (CRITICAL present)
tc_pass_rate    = 0.25   (staged payload)

confidence = 0.3×0.9 + 0.3×0.0 + 0.4×0.25
           = 0.27 + 0.00 + 0.10
           = 0.37  ✅
```

PR comment posted:

```
❌ Confidence: 0.37 — CRITICAL security finding
   TC pass rate: 25% · Critic: iteration 2 · 3 FRAGILE tests flagged
   Security: 1 CRITICAL (hardcoded secret, line 14) · 2 warnings
   Requirements: REQ-001 ✅ REQ-002 ✅ REQ-003 ❌ (no tests generated)
   Maestro case opened: CASE-00142
```

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15. Tech Stack

Layer	Technology
AI agents	Anthropic Python SDK `anthropic>=0.40.0`
Models	`claude-haiku-4-5-20251001` (triage), `claude-sonnet-4-6` (writer/critic/security), `claude-opus-4-7` (optional)
Static analysis	Semgrep via `asyncio.create_subprocess_exec` + tempfile (p/python, p/owasp-top-ten, p/secrets)
Test runner	pytest + `--junitxml` via `asyncio.create_subprocess_exec`
TC integration	JUnit XML → UiPath Test Cloud REST API v2 via `httpx.AsyncClient`
Webhook server	FastAPI + uvicorn (HMAC-SHA256 verified, `asyncio.Queue` background worker)
Shared state	SQLite (FastAPI writes, Streamlit polls at 500ms)
Dashboard	Streamlit with 9-step pipeline indicator
Maestro	UiPath Maestro CRITICAL escalation (graceful skip if unavailable)
Containers	Docker + docker-compose (shared SQLite volume)
Language	Python 3.11+ in `.venv`
Key management	python-dotenv, fully annotated `.env.example`, `TC_DRY_RUN` dev flag

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16. Devpost Submission Notes

- **Track:** Track 3 — Test Cloud
- **Tagline:** "Three agents. One adversarial loop. Zero untested edge cases."
- **Clarify:** Writer generates Python *test* code (pytest), not implementation code
- **TC contract:** JUnit XML — explain explicitly so judges don't assume raw Python is posted
- **HMAC:** Call it out — a security project that verifies its own webhook is internally consistent
- **Real API gotcha (include verbatim):**
  ⚠️ During integration we found that GitHub's webhook payload includes `changed_files` as an integer (the count), not an array. Fetching the actual file list requires a separate call to `GET /repos/{owner}/{repo}/pulls/{number}/files`. We handle this explicitly — the kind of detail that only shows up when you actually integrate, not just read the docs.
- `TC_DRY_RUN` **flag:** Mention it — signals professional dev hygiene
- **Maestro:** One paragraph, one screenshot, "cross-platform escalation bonus"
- **Badge:** "Built with Claude Code" in README + Devpost body